# Groundwater Outlook of Bengaluru City – Interim Report -II - April 2025 Projections

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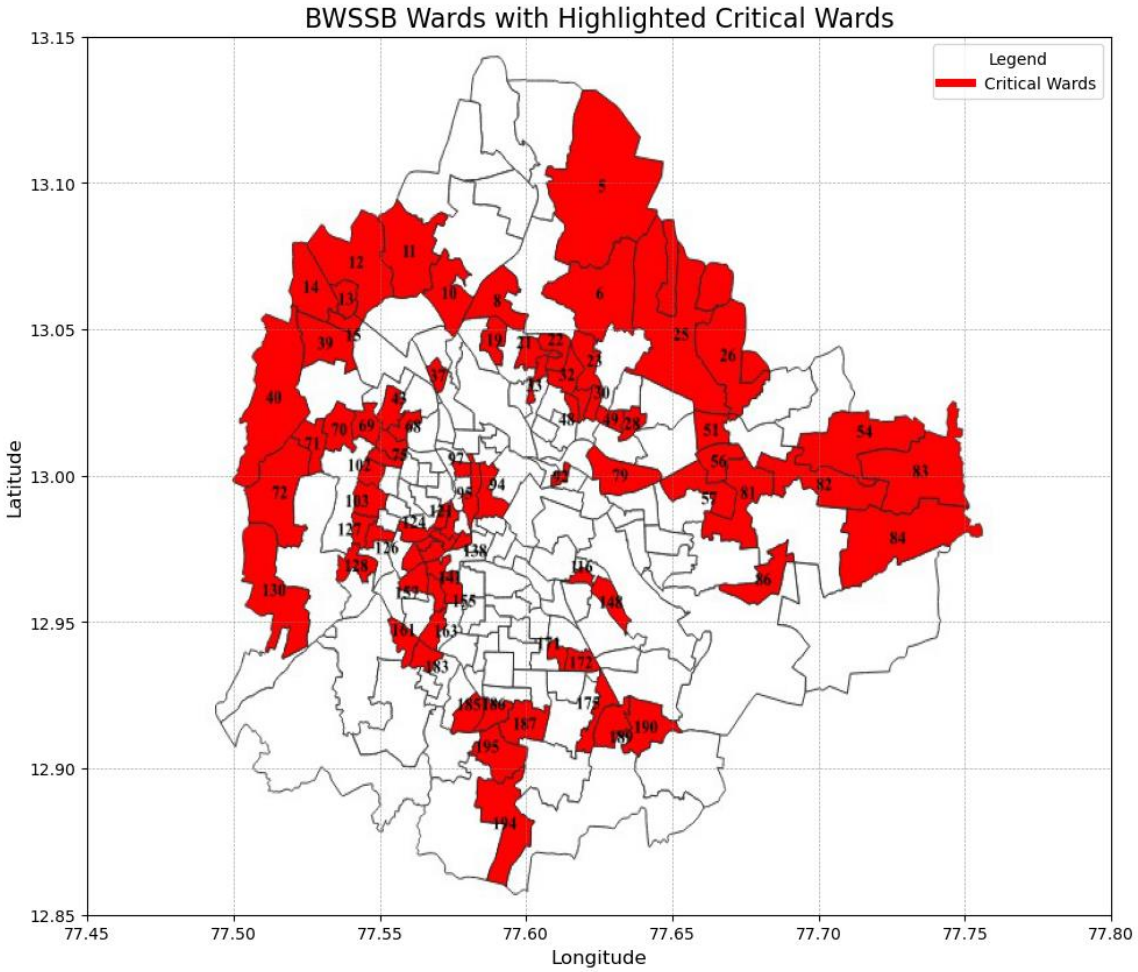
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## Introduction

This report presents a comprehensive analysis of Bangalore's groundwater situation, emphasizing the wards at high risk of water scarcity during summer. The study combines geospatial, historical, and recent data to identify critical areas, predict trends in groundwater levels, and propose actionable recommendations for sustainable water management.

## Critical Wards Map

Below is a visual representation of the 80 critical wards identified as at risk of severe water scarcity:




## Objectives

1. Identify Bangalore's wards at high risk of water scarcity.

2. Provide actionable insights and targeted recommendations.

3. Develop a groundwater outlook dashboard for visualization and decision-making.

## Summary of Findings

### Critical Wards Identified

- 80 critical wards (including 110 villages) were identified as highly groundwater-dependent and at risk of severe shortages. The list of the 80 critical wards is given in Table 1 of the report.

**Predicted Water Level Drops by April 2025 (from Dec 2024 levels):**

Core Area: 5 meters
CMC Area: 10-15 meters
110 Villages: 20-25 meters

### Groundwater Extraction Trends

- Current extraction levels: Approximately 800 million liters per day (MLD).
- Peripheral areas show the highest dependency on borewells.

### Projected Scenarios

- South-East Bangalore and Whitefield areas face the highest stress.
- Continued extraction without intervention will exacerbate shortages in critical wards.

## Methodology

### Data Sources

1. Borewell Data: Comprehensive data from private and BWSSB-operated borewells.
2. Water Consumption Trends: Last five years of per capita consumption data (LPCD).
3. Geospatial Information: Ward-level shapefiles.
4. Groundwater Levels: Historical (10 years) and recent (2 years) data from Central Ground Water Board (CGWB).

### Analytical Approach

1. Geospatial Mapping:
- Ward shapefiles were used to map borewell density and groundwater dependence.
- Critical zones were identified by analyzing the data for borewell usage patterns.

2. Groundwater Trend Analysis:
- Historical groundwater levels from CGWB were analyzed to predict seasonal trends.
- Recent data validated these trends and improved projections for 2025.

3. Data Analysis and Parameter Extraction:
- Available data was analyzed comprehensively to derive various parameters related to groundwater dependency and risks.
- This approach helped in identifying 80 critical wards at risk of severe shortages.

4. Consumption and Extraction Trends:
- LPCD consumption trends were extracted and compared across wards to identify high-demand zones.

5. Urban Challenges:
- Challenges such as limited pipeline access, neglected lake maintenance, and overexploitation of groundwater were incorporated based on the WELL Labs report.

6. Predictive Modeling:
- Time-series modeling techniques were used to predict groundwater depletion rates for 2025.

## Field Surveys

Ongoing manual surveys in critical wards in collaboration with Dayananda Sagar Institute are capturing granular data to refine ward-wise variability in groundwater usage and recharge potential.

## Key Conclusions

### Water Scarcity Risks

- Severe shortages are anticipated in wards with high borewell density and groundwater dependency.
- South-East Bangalore and peripheral areas near Whitefield face the highest levels of stress.

### Recharge Opportunities

- Areas with high recharge potential have been identified through detailed analysis of available data.
- These zones can play a critical role in groundwater replenishment if properly managed.

### Infrastructure Gaps

- Real-time monitoring infrastructure (e.g., piezometers) is insufficient in critical wards, delaying timely interventions.

## Recommendations

### Enhance Water Supply Infrastructure

1. Cauvery Connect Stations (CCS):
- Establish CCS near critical wards to provide alternative water sources.
- Each station should serve a 5-km radius.

2. Lake Recharge:
- Fill lakes in eastern and northern Bangalore with treated wastewater.
- Regularly monitor lake water quality to ensure effective recharge.

### Demand-Side Measures

- Sensitize apartment complexes to reduce dependency on borewells and switch to Cauvery water.
- Enforce stricter monitoring of high-extraction zones.

### Monitoring and Management

- Install piezometers in critical wards for real-time groundwater level monitoring.
- Regularly assess lake water quality and its impact on recharge rates.

**Stakeholder Engagement**

- Involve local communities in participatory groundwater management to improve compliance and sustainability.

## Next Steps

**Ongoing Activities**

1. Completion of high-frequency data analysis to evaluate the effects of high pumping.
2. Refinement of groundwater projections using updated field survey data.

**Future Goals**

- Development of a web-enabled dashboard to visualize groundwater trends, risks, and actionable areas.
- Integration of participatory groundwater management strategies into policy frameworks.

## Critical Wards Table

The following table lists 80 critical wards identified during the analysis.

Table 1: List of Critical Wards

| Sl No | WARD NO | WARD NAME | LATITUDE | LONGITUDE |
|---|---|---|---|---|
| 1 | 5 | Jakkuru | 13.09625029 | 77.62331396 |
| 2 | 6 | Thanisandra | 13.05652486 | 77.62241074 |
| 3 | 8 | Kodigehalli | 13.0538456 | 77.57908776 |
| 4 | 10 | Dodda Bommasandra | 13.05659513 | 77.5586925 |
| 5 | 11 | Kuvempu Nagar | 13.07207982 | 77.54186935 |
| 6 | 12 | Shettihalli | 13.06799669 | 77.51940919 |
| 7 | 13 | Mallasandra | 13.05443636 | 77.51512592 |
| 8 | 14 | Bagalakunte | 13.0588925 | 77.50028001 |
| 9 | 15 | T Dasarahalli | 13.04535802 | 77.51473575 |
| 10 | 19 | Sanjaya Nagar | 13.03930133 | 77.57779971 |
| 11 | 21 | Hebbala | 13.03405446 | 77.59301874 |
| 12 | 22 | Vishwanath Nagenahalli | 13.03517448 | 77.60374058 |
| 13 | 23 | Nagavara | 13.03150907 | 77.61669583 |
| 14 | 25 | Horamavu | 13.04456134 | 77.65327081 |
| 15 | 26 | Ramamurthy Nagar | 13.03361313 | 77.67653867 |
| 16 | 28 | Kammanahalli | 13.0083737 | 77.63599273 |
| 17 | 30 | Kadugondanahalli | 13.01554351 | 77.61986406 |
| 18 | 31 | Kushal Nagar | 13.01627331 | 77.61501741 |

| | | | | |
|---|---|---|---|---|
| **19** | **32** | **Kaval Bairasandra** | **13.02635057** | **77.608569** |
| **20** | **33** | **Manorayanapalya** | **13.0267427** | **77.59730529** |
| **21** | **37** | **Yeshwanthpura** | **13.026029** | **77.55385677** |
| **22** | **39** | **Chokkasandra** | **13.03805041** | **77.50630669** |
| **23** | **40** | **Dodda Bidarakallu** | **13.01807587** | **77.4845431** |
| **24** | **43** | **Nandini Layout** | **13.01344994** | **77.53414161** |
| **25** | **48** | **Muneshwara Nagar** | **13.01416004** | **77.61097854** |
| **26** | **49** | **Lingarajapura** | **13.00994594** | **77.62697185** |
| **27** | **51** | **Vijnanapura** | **13.00606326** | **77.6695653** |
| **28** | **54** | **Hudi** | **13.02237552** | **77.70549333** |
| **29** | **56** | **A Narayanapura** | **12.99447441** | **77.67258266** |
| **30** | **57** | **C V Raman Nagar** | **12.9839497** | **77.66520285** |
| **31** | **68** | **Mahalakshimpuram** | **13.00734965** | **77.54004259** |
| **32** | **69** | **Laggere** | **13.00768736** | **77.52389999** |
| **33** | **70** | **Rajagopal Nagar** | **13.00619961** | **77.51224925** |
| **34** | **71** | **Hegganahalli** | **13.00100553** | **77.50097544** |
| **35** | **72** | **Herohalli** | **12.98280727** | **77.48715854** |
| **36** | **74** | **Shakthi Ganapathi Nagar** | **12.99504803** | **77.53158397** |
| **37** | **75** | **Shankar Matt** | **12.99998428** | **77.53440777** |
| **38** | **79** | **Sarvagna Nagar** | **12.98917003** | **77.63103948** |
| **39** | **81** | **Vijnana Nagar** | **12.97849298** | **77.68177013** |
| **40** | **82** | **Garudachar Playa** | **12.98590064** | **77.71728282** |
| **41** | **83** | **Kadugodi** | **12.99068773** | **77.7579089** |
| **42** | **84** | **Hagadur** | **12.9660702** | **77.74836474** |
| **43** | **86** | **Marathahalli** | **12.95074306** | **77.69149524** |
| **44** | **92** | **Shivaji Nagar** | **12.98870168** | **77.60587284** |
| **45** | **94** | **Gandhinagar** | **12.98156481** | **77.57566501** |
| **46** | **95** | **Subhash Nagar** | **12.9826668** | **77.56862746** |
| **47** | **97** | **Dayananda Nagar** | **12.99109706** | **77.56419467** |
| **48** | **102** | **Vrisabhavathi Nagar** | **12.98977242** | **77.52587903** |
| **49** | **103** | **Kaveripura** | **12.97974227** | **77.52489788** |
| **50** | **116** | **Nilasandra** | **12.95115994** | **77.61464413** |
| **51** | **121** | **Binnipete** | **12.97179842** | **77.55909691** |

| | | | | |
|---|---|---|---|---|
| **52** | **122** | **Kempapura Agrahara** | **12.97231262** | **77.55547938** |
| **53** | **124** | **Hosahalli** | **12.96766915** | **77.54381199** |
| **54** | **126** | **Maruthi Mandir ward** | **12.96671055** | **77.52846435** |
| **55** | **127** | **Mudalapalya** | **12.96923922** | **77.52164058** |
| **56** | **128** | **Nagarabhavi** | **12.95584273** | **77.51936615** |
| **57** | **130** | **Ullalu** | **12.94671614** | **77.48461819** |
| **58** | **134** | **Bapuji Nagar** | **12.9579761** | **77.54322141** |
| **59** | **135** | **Padarayanapura** | **12.96370255** | **77.54881205** |
| **60** | **136** | **Jagajivanaramnagar** | **12.96054475** | **77.55194145** |
| **61** | **137** | **Rayapuram** | **12.96498808** | **77.55637818** |
| **62** | **138** | **Chalavadipalya** | **12.96457932** | **77.56444823** |
| **63** | **141** | **Azad Nagar** | **12.95471714** | **77.5540273** |
| **64** | **148** | **Ejipura** | **12.94231592** | **77.62703361** |
| **65** | **155** | **Hanumanth Nagar** | **12.9471215** | **77.55988387** |
| **66** | **156** | **Srinagar** | **12.94532753** | **77.55349437** |
| **67** | **157** | **Gali Anjenaya Temple ward** | **12.95015925** | **77.54465603** |
| **68** | **161** | **Hosakerehalli** | **12.92779693** | **77.53950583** |
| **69** | **163** | **Katriguppe** | **12.93164146** | **77.55222329** |
| **70** | **171** | **Gurappanapalya** | **12.9228664** | **77.60461515** |
| **71** | **172** | **Madivala** | **12.92001823** | **77.61441839** |
| **72** | **175** | **Bommanahalli** | **12.90074111** | **77.62121176** |
| **73** | **183** | **Chikkalsandra** | **12.92168479** | **77.5484074** |
| **74** | **185** | **Yelchenahalli** | **12.90022868** | **77.56701407** |
| **75** | **186** | **Jaraganahalli** | **12.90042089** | **77.57748806** |
| **76** | **187** | **Puttenahalli** | **12.89731775** | **77.59060305** |
| **77** | **189** | **Hongasandra** | **12.89676889** | **77.62751717** |
| **78** | **190** | **Mangammanapalya** | **12.89616674** | **77.64182309** |
| **79** | **194** | **Gottigere** | **12.86059126** | **77.58213245** |
| **80** | **195** | **Konankunte** | **12.88577483** | **77.57990145** |

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